

4.10.16

EE25BTECH11014 - Bhoomika Lokesh

Question: Solve the given system of linear equations:

$$x - 3y - 7 = 0 \quad (1)$$

$$3x - 3y - 15 = 0 \quad (2)$$

Solution:

The given equations of line can be expressed as:

$$\mathbf{n}^T \mathbf{x} = c$$

$$\begin{pmatrix} 1 & -3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 7 \quad (3)$$

$$\begin{pmatrix} 3 & -3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 15 \quad (4)$$

The matrix we get by the above equations is as follows:

$$\begin{pmatrix} 1 & -3 \\ 3 & -3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 7 \\ 15 \end{pmatrix} \quad (5)$$

The following is the augmented matrix which can be solved using Gaussian elimination.

$$\left(\begin{array}{cc|c} 1 & -3 & 7 \\ 3 & -3 & 15 \end{array} \right) \xrightarrow{R_2 \rightarrow R_2 - 3R_1} \left(\begin{array}{cc|c} 1 & -3 & 7 \\ 0 & 6 & -6 \end{array} \right) \quad (6)$$

$$\left(\begin{array}{cc|c} 1 & -3 & 7 \\ 0 & 6 & -6 \end{array} \right) \xrightarrow{R_2 \rightarrow \frac{R_2}{6}} \left(\begin{array}{cc|c} 1 & -3 & 7 \\ 0 & 1 & -1 \end{array} \right) \quad (7)$$

$$\left(\begin{array}{cc|c} 1 & -3 & 7 \\ 0 & 1 & -1 \end{array} \right) \xrightarrow{R_1 \rightarrow R_1 + 3R_2} \left(\begin{array}{cc|c} 1 & 0 & 4 \\ 0 & 1 & -1 \end{array} \right) \quad (8)$$

From the matrix,

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 4 \\ -1 \end{pmatrix} \quad (9)$$

$$\therefore \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 4 \\ -1 \end{pmatrix} \quad (10)$$

Hence the system of equations has a unique solution, $x = 4$ and $y = -1$.

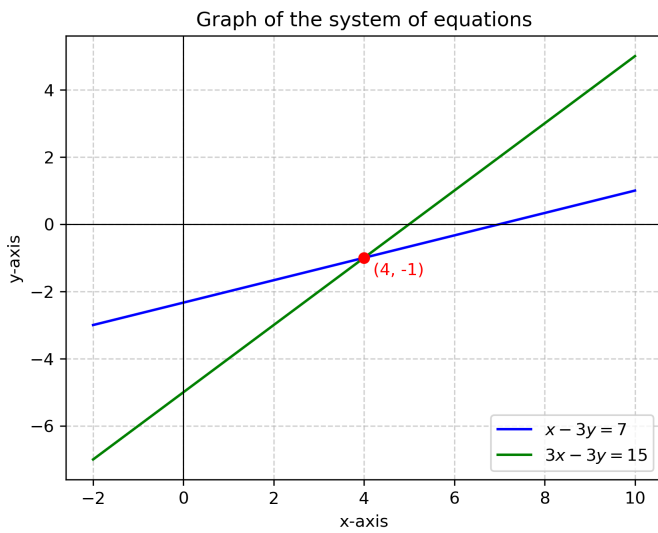


Fig. 0.1: System of Linear Equations